

Basic Input/Output in Python

Using `print()`, `input()`, type conversion, and f-strings

Introductory Python Programming

Lesson roadmap

- 1 Displaying data with `print()`
- 2 Receiving keyboard input with `input()`
- 3 Converting input data types
- 4 Input → Process → Output
- 5 Formatting output with f-strings
- 6 Common input/output errors
- 7 Integrated practice exercises

Learning objectives

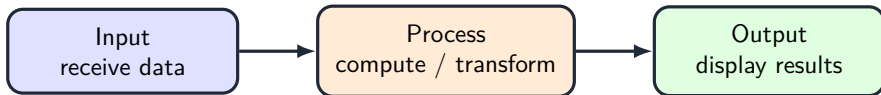
After this lesson, students will be able to:

- use `print()` to display data;
- print multiple values with `sep` and `end`;
- use `input()` to receive keyboard input;
- explain why `input()` returns `str`;
- convert input to `int` or `float`;
- combine input, processing, and output in one program;
- format output using basic f-strings.

Lesson scope

All programs are written directly with basic statements. User-defined functions (`def`) are not used yet.

The basic program pattern



Example: rectangle area

Input length and width → calculate area → display the result.

Displaying data with print()

Basic syntax

```
print(value)
```

Example

```
print("Hello, Python!")
```

Output

```
Hello, Python!
```

Printing numbers and variables

Numbers

```
print(100)  
print(3.14)
```

Variables

```
name = "An"  
age = 18  
print(name)  
print(age)
```

Key idea

`print()` can display literal values and values stored in variables.

Printing multiple values

`print()` can receive more than one value.

```
name = "An"  
age = 18  
print("Name:", name)  
print("Age:", age)
```

Output

```
Name: An  
Age: 18
```

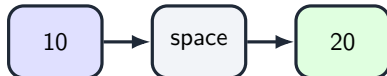
- By default, Python inserts a space between printed values.

Default separator

```
x = 10  
y = 20  
print(x, y)
```

Output

10 20



The sep parameter

Purpose

sep controls what is placed between printed values.

```
print("2026", "09", "06", sep="-")
```

2026-09-06

```
print("Python", "Java", "C++", sep=" | ")
```

Python | Java | C++

The end parameter

Default behavior

After each `print()` statement, Python moves to a new line.

```
print("Hello")  
print("Python")
```

Hello
Python

```
print("Hello", end=" ")  
print("Python")
```

Hello Python

Self-Check 1

Question 1

What is the output?

```
print("A", "B", "C", sep="-")
```

Question 2

What is the output?

```
print("Hello", end=" ")  
print("World")
```

Answers to Self-Check 1

- ① A-B-C
- ② Hello World

Receiving input with input()

Syntax

```
variable = input("Prompt")
```

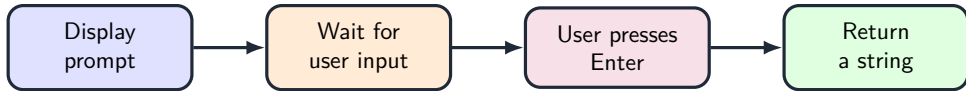
Example

```
name = input("Enter your name: ")  
print("Hello", name)
```

Meaning

The prompt is shown to the user. The program waits until the user types data and presses Enter.

What does `input()` do?



Four steps

Display a prompt, wait for input, receive the entered text, and return the text to the program.

`input()` returns a string

```
age = input("Enter your age: ")  
print(type(age))
```

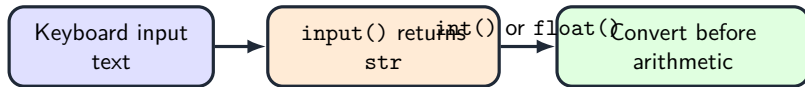
If the user enters 18, the result is still:

```
<class 'str'>
```

Important

Even when the input looks like a number, `input()` returns data of type `str`.

Why type conversion is needed



Rule

When input data is used in arithmetic calculations, convert it to the appropriate numeric type.

Converting to an integer with `int()`

```
age = int(input("Enter your age: "))  
print(age + 1)
```

If the user enters 18, the output is:

19

- Use `int()` for whole numbers.
- Examples: age, quantity, number of students.

Converting to a floating-point number with `float()`

```
price = float(input("Enter price: "))  
print(price)
```

- Use `float()` for numbers that may contain decimals.
- Examples: price, height, weight, temperature.

Without type conversion

```
x = input("Enter x: ")  
y = input("Enter y: ")  
print(x + y)
```

If the user enters 10 and 20, the output is:

1020

Why?

x and y are strings, so + concatenates them.

With type conversion

```
x = int(input("Enter x: "))  
y = int(input("Enter y: "))  
print(x + y)
```

Output:

30

Difference

After conversion, x and y are integers, so + performs addition.

Self-Check 2

Question 1

What does this code print if the user enters 5?

```
x = input("x = ")  
print(x * 3)
```

Question 2

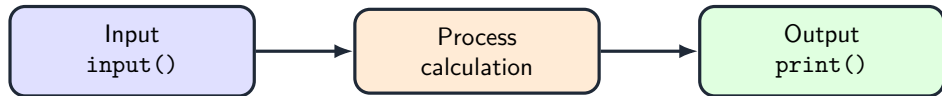
How can the code be modified so that the result is 15?

Answers to Self-Check 2

- 1 The output is 555. The value is the string "5", so multiplying by 3 repeats the string.
- 2 Convert the input to an integer:

```
x = int(input("x = "))  
print(x * 3)
```

The Input - Process - Output model



Programming habit

For beginner programs, clearly identify input, processing, and output before coding.

Example: rectangle area

```
length = float(input("Length: "))  
width = float(input("Width: "))  
area = length * width  
print("Area:", area)
```

Input	Process	Output
length, width	$\text{length} * \text{width}$	<code>print(...)</code>

Formatting output with f-strings

F-strings provide a clear way to combine text with variable values.

```
name = "An"  
age = 18  
print(f"My name is {name}.")  
print(f"I am {age} years old.")
```

Output

```
My name is An.  
I am 18 years old.
```

Expressions inside f-strings

```
x = 10  
y = 5  
print(f"{x} + {y} = {x + y}")
```

Output:

10 + 5 = 15

Key idea

Expressions inside { } are evaluated before being inserted into the string.

Formatting floating-point numbers

```
price = 19.5678  
print(f"Price: {price:.2f}")
```

Output:

Price: 19.57

Meaning

.2f displays a floating-point number with 2 digits after the decimal point.

Common error: forgetting type conversion

Incorrect

```
age = input("Age: ")  
next_age = age + 1
```

Correct

```
age = int(input("Age: "))  
next_age = age + 1
```

Reason

age is a string, while 1 is an integer. They cannot be added directly.

Common error: invalid numeric input

```
age = int(input("Age: "))
```

Problem

If the user enters `eighteen`, Python cannot convert that string into an integer.

Lesson scope

For now, assume that users enter data in the required format. Input validation will be introduced later.

Common error: variable vs string literal

Print the text x

```
x = 10  
print("x")
```

Output:

x

Print the value of x

```
x = 10  
print(x)
```

Output:

10

Integrated example 1: personal information

```
name = input("Name: ")
age = int(input("Age: "))

print(f"Hello {name}!")
print(f"Next year you will be {age + 1}.")
```

IPO analysis

- Input: name and age
- Process: compute age + 1
- Output: greeting and next year's age

Integrated example 2: total cost

```
price = float(input("Product price: "))
quantity = int(input("Quantity: "))
total = price * quantity

print(f"Total: {total:.2f}")
```

Important conversions

- price: use float()
- quantity: use int()
- total: display with .2f

Integrated example 3: Celsius to Fahrenheit

Formula:

$$F = \frac{9}{5}C + 32$$

```
celsius = float(input("Celsius: "))  
fahrenheit = 9 / 5 * celsius + 32  
print(f"Fahrenheit: {fahrenheit:.2f}")
```

Predict the output: exercises

Exercise 1

```
x = "10"  
y = "5"  
print(x + y)
```

Exercise 2

```
x = 10  
y = 5  
print("Result:", x + y)
```

Exercise 3

```
print("A", "B", sep=":", end=" ")  
print("C")
```

Exercise 4

```
x = 7  
print(f"x = {x}, x^2 = {x * x}")
```

Answers: predict the output

- ① 105
- ② Result: 15
- ③ A:B C
- ④ $x = 7$, $x^2 = 49$

Practice exercises 1-4

General requirement

Write statements directly using the Input - Process - Output model. Do not use def yet.

- 1 Greeting: input a name and display Hello, <name>!
- 2 Age next year: input age and display next year's age.
- 3 Sum and product: input two integers, then display their sum and product.
- 4 Rectangle: input length and width, then display area and perimeter.

Practice exercises 5-8

- ① Total purchase cost: input product name, unit price, and quantity, then display an invoice.
- ② Temperature conversion: convert Celsius to Fahrenheit and display 2 decimals.
- ③ Average score: input three scores and display the average with 2 decimals.
- ④ Convert seconds: input total seconds, then display full minutes and remaining seconds.

Useful operators for Exercise 8

// for integer division and % for remainder.

Integrated practice: trip cost

Input

Distance traveled, fuel consumption in liters per 100 km, and fuel price.

Formulas

$$\text{fuel_used} = \frac{\text{distance} \times \text{consumption}}{100}$$

$$\text{cost} = \text{fuel_used} \times \text{fuel_price}$$

Integrated practice: sample output

```
Distance: 250  
Consumption (L/100km): 7.5  
Fuel price: 23000  
Fuel used: 18.75 L  
Estimated cost: 431250.00
```

Focus

This exercise combines input, type conversion, arithmetic, and formatted output.

Self-assessment after the lesson

- I know how to use `print()`.
- I understand `sep` and `end`.
- I know how to use `input()`.
- I remember that `input()` returns `str`.
- I know when to use `int()` or `float()`.
- I can write a program using the Input - Process - Output model.
- I can format floating-point values using `.2f`.
- I can independently write simple programs without using `def`.

Knowledge summary

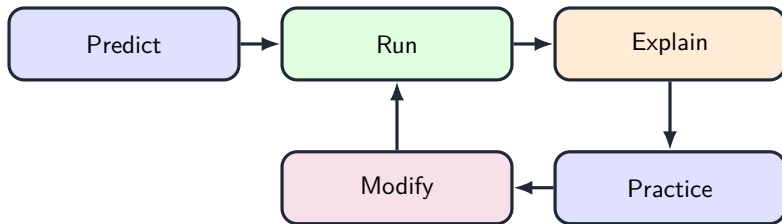
Important tools:

```
print(...)
input(...)
int(...)
float(...)
type(...)
```

Common patterns:

```
age = int(input("Age: "))
price = float(input("Price: "))
print(f"Result: {result:.2f}")
```

Final takeaway



Learning process

Predict → Run → Explain → Modify → Practice.